Intro (Creative, with Innovations Highlighted)

Meet the **Smart Multilingual Meetings & Lectures Assistant**—a next-gen, AI-powered tool designed for the world's knowledge workers, students, and teams.
Upload any meeting or lecture (audio or video), and instantly unlock time-synced transcripts, smart summaries, and deep, conversational Q&A—in 15 different languages.

Key Innovations & Unique Features:

- **Handles Long Meetings & Lectures:**
  If your audio/video is too lengthy for the AI's token limit, our system **automatically condenses transcripts** into focused, concise segments—so you never lose key information, even from the longest recordings.
- **AI that speaks your language:**
  Provides accurate summaries and enables natural language conversation (Q&A) in **15+ languages**, making global collaboration effortless.
- **Intelligent Prompt Engineering:**
  We didn't just "call OpenAI"—every AI task uses **custom-engineered prompts(on which we experimented as lot to choose the most effective prompts)** for summarization, semantic search, and safe, respectful output.
  added prompts to never catch or use Abusive words of any language, ensuring clean and professional results in every language.
- **Blazing Fast, Runs Anywhere:**
  Packaged with **Docker** for effortless, one-command deployment on any machine (CPU or GPU supported), so you can scale up or down as needed very easily and minimal setup instructions.
- **Lightning-Fast Speech-to-Text:**
  Utilizes the advanced **Whisper model** to transcribe audio and video, automatically detects CPU and GPU for maximum performance—even on complex, multi-language content.
- **Modern, Modular Frontend (React):**
  Built with React using modular, reusable components for rapid development and easy maintenance. This architecture ensures the UI is smooth, responsive, fast and highly effective, delivering a seamless experience for users.
- **Enterprise-Ready Architecture:**
  Uses Nginx reverse proxies to securely route user requests to the frontend Docker container; any API or dynamic requests from the frontend are automatically redirected by Nginx to the backend Docker container. This enables secure, modular, and scalable client-server communication, while simplifying deployment and management.
- **Scalable by Design:**
  Async Endpoints: Utilized FastAPI's asynchronous API routes for efficient, concurrent processing of uploads, summaries, and chat queries.

StreamingResponse: Utilized FastAPI's StreamingResponse to deliver chatbot answers in real time, efficiently streaming large outputs in small chunks—so even long transcripts are handled smoothly without extra memory usage.

Tech at a Glance

- **Frontend:** React (modular, component-based), hosted separately for max speed and reliability
- **Backend:** Python (FastAPI), Dockerized, scalable, GPU/CPU support
- **Speech-to-Text:** Whisper model (fast, accurate, supports all hardware)
- **AI Summaries & Q&A:** OpenAI GPT-3.5 Turbo with prompt engineering

- **Infrastructure:** Docker, Nginx (reverse proxy), modular microservice architecture